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B.Tech. Degree V Semester Regular Examination November 2021

ME 19-205-0502 MECHANICS OF MACHINERY

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer **ALL** questions)

(8 × 3 = 24)

- I. (a) Draw the sketch of a mechanism in which a point traces an exact straight line. The mechanism must be made of only revolute pairs. Prove that the point traces an exact straight line motion.
- (b) The angle between the axes of two shafts connected by Hooke's joint is 18° . Determine the angle turned through by the driving shaft when the velocity ratio is maximum and unity.
- (c) What is the difference between involute and cycloidal teeth profile.
- (d) Explain briefly the differences between simple, compound, and epicyclic gear trains. What are the special advantages of epicyclic gear trains?
- (e) Draw the displacement, velocity and acceleration diagrams for a follower when it moves with simple harmonic motion. Derive the expression for velocity and acceleration during outstroke and return stroke of the follower.
- (f) A body, resting on a rough horizontal plane required a pull of 180 N inclined at 30° to the plane just to move it. It was found that a push of 220 N inclined at 30° to the plane just moved the body. Determine the weight of the body and the coefficient of friction.
- (g) Describe the construction and operation of a rope brake absorption dynamometer.
- (h) Explain Freudenstein's method of three point synthesis of mechanisms.

PART B

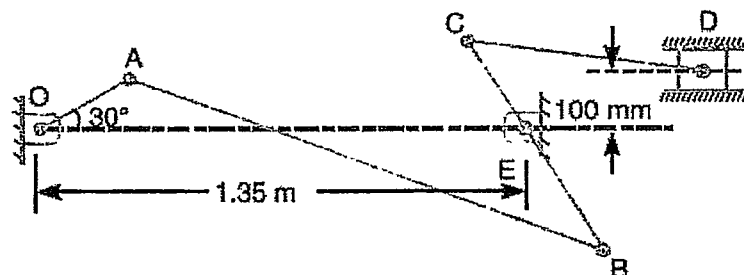
(4 × 12 = 48)

- II. A mechanism, as shown in the figure, has the following dimensions: (12)

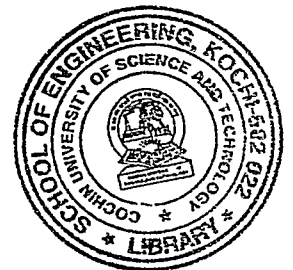
OA = 200 mm; AB = 1.5 m; BC = 600 mm; CD = 500 mm; BE = 400 mm

Locate all the instantaneous centres. If crank OA rotates uniformly at 120 r.p.m. clockwise, find:

- (i) The velocity of B, C and D.
- (ii) The angular velocity of the links AB, BC and CD.



OR



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- III. In a four bar chain ABCD, link AD is fixed and the crank AB rotates at 10 radians per second clockwise. Lengths of the links are AB = 60 mm; BC = CD = 70 mm; DA = 120 mm. When angle DAB = 60° and both B and C lie on the same side of AD, find: (12)
- Angular velocities (magnitude and direction) of BC and CD.
 - Angular acceleration of BC and CD.
- IV. (a) A 20° involute pinion with 20 teeth drives a gear having 60 teeth. Module is 8 mm and addendum of each gear is 10 mm. (6)
- State whether interference occurs or not. Give reasons.
 - Find the length of path of approach and arc of approach if pinion is the driver.
- (b) Two mating gears have 20 and 40 involute teeth of module 10 mm and 20° pressure angle. The addendum on each wheel is to be made of such a length that the line of contact on each side of the pitch point has half the maximum possible length. Determine the addendum height for each gear wheel, length of the path of contact, arc of contact and contact ratio. (6)
- OR**
- V. (a) In an epicyclic gear train, the internal wheels A and B and compound wheels C and D rotate independently about axis O. The wheels E and F rotate on pins fixed to the arm G. E gears with A and C and F gears with B and D. All the wheels have the same module and the number of teeth are : $T_C = 28$; $T_D = 26$; $T_E = T_F = 18$. (12)
- Sketch the arrangement.
 - Find the number of teeth on A and B.
 - If the arm G makes 100 r.p.m. clockwise and A is fixed, find the speed of B.
 - If the arm G makes 100 r.p.m. clockwise and wheel A makes 10 r.p.m. counter clockwise, find the speed of wheel B.
- VI. A cam rotating clockwise at a uniform speed of 200 r.p.m. is required to move an offset roller follower with a uniform and equal acceleration and retardation on both the outward and return strokes. The angle of ascent, the angle of dwell (between ascent and descent) and the angle of descent is 120°, 60° and 90° respectively. The follower dwells for the rest of cam rotation. The least radius of the cam is 50 mm, the lift of the follower is 25 mm and the diameter of the roller is 10 mm. The line of stroke of the follower is offset by 20 mm from the axis of the cam. Draw the cam profile and find the maximum velocity and acceleration of the follower during the outstroke. (12)
- OR**
- VII. (a) The pitch of 50 mm mean diameter threaded screw of a screw jack is 12.5 mm. The coefficient of friction between the screw and the nut is 0.13. Determine the torque required on the screw to raise a load of 25 kN, assuming the load to rotate with the screw. Determine the ratio of the torque required to raise the load to the torque required to lower the load and also the efficiency of the machine. (6)
- (b) A thrust shaft of a ship has 6 collars of 600 mm external diameter and 300 mm internal diameter. The total thrust from the propeller is 100 kN. If the coefficient of friction is 0.12 and speed of the engine 90 r.p.m., find the power absorbed in friction at the thrust block, assuming: (6)
- Uniform pressure.
 - Uniform wear.

(Continued to 3)

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VIII. (a) Establish a formula for the maximum torque transmitted by a single plate clutch of external and internal radii r_1 and r_2 , if the limiting coefficient of friction is μ and the axial spring load is W . Assume that the pressure intensity on the contact faces is uniform. (4)

(b) A centrifugal friction clutch has a driving member consisting of a spider carrying four shoes which are kept from contact with the clutch case by means of flat springs until increase of centrifugal force overcomes the resistance of the springs and the power is transmitted by friction between the shoes and the case. Determine the necessary mass of each shoe if 22.5 kW is to be transmitted at 750 r.p.m. with engagement beginning at 75% of the running speed. The inside diameter of the drum is 300 mm and the radial distance of the centre of gravity of each shoe from the shaft axis is 125 mm. Assume $\mu = 0.25$. (8)

OR

IX. Synthesize a four-bar mechanism to generate a function $y = \sin x$ for $0 \leq x \leq 90^\circ$. The range of the output crank may be chosen as 60° while that of input crank be 120° . Assume three precision points which are to be obtained from Chebyshev spacing. Assume fixed link to be 52.5 mm long and $\theta_1 = 105^\circ$ and $\phi_1 = 66^\circ$. (12)

